Debiasing Technique Rankings Are Metric- and Domain-Dependent: A Cross-Domain Exploration of Anchoring Bias Interventions in LLMs

Tom Howard*

February 2026

Abstract

Do debiasing techniques for anchoring bias transfer across domains? We evaluate five techniques—Devil's Advocate, SACD (Self-Ask Chain of Deliberation), Premortem, Random Control, and Outside View (excluded from rankings due to confound)—across 21,139 trials spanning six domains: salary negotiation, loan approval, medical triage, and three judicial sentencing scenarios.

Our core finding: our exploratory cross-domain analysis (4 models) suggests no technique consistently outperforms. Technique rankings vary by domain, and bootstrap 95% confidence intervals overlap for all pairwise #1 vs #2 comparisons—we cannot statistically distinguish which technique is "best" in any domain. Point estimates suggest SACD performs well for salary (12.6% MAD, rank #2) but underperforms in fraud (46.2% MAD, rank #3), though these differences are not statistically significant. Random Control—which provides no debiasing content, only additional conversation turns—ranks #1 in 4 of 6 domains by point estimate.

We apply Mean Absolute Deviation (MAD) from unanchored baseline as a complementary metric that captures per-trial error obscured by aggregate measures. SACD's aggregate 93.7% of baseline in judicial sentencing masks 18.1% per-trial deviation due to bidirectional errors that cancel out.

A deep-dive validation on judicial sentencing (14,152 trials, 10 models) confirms substantial model-specific variance: the same technique can improve one model's responses while degrading another's. Combined with domain dependence, this implies practitioners must test debiasing interventions per-model AND per-task rather than relying on published rankings.

1 Introduction

Human debiasing techniques—Devil's Advocate, Premortem, deliberative reasoning—are increasingly applied to LLMs. But do they transfer across tasks? A technique that works for salary negotiation may fail for medical triage. Prior evaluations typically test single domains; we test six.

We report findings from 21,139 trials across six domains (salary, loan, medical, three judicial scenarios) using models spanning capability tiers (Opus 4.6, Sonnet 4.6, Haiku 4.5, GPT-5.2). Our core finding: **technique rankings vary by domain, and we cannot identify a universally**

*Correspondence: tom@voder.ai. GitHub: @tompahoward. This research was conducted with AI assistance; see AI Assistance Disclosure.

superior approach. In the cross-domain analysis (4 models), bootstrap 95% CIs overlap for all #1 vs #2 comparisons—we cannot statistically distinguish which technique is "best" in any domain.

We also identify a methodological flaw in prior work: **susceptibility**—measuring whether responses vary with anchors—conflates consistency with correctness. A technique can show low susceptibility by being consistently wrong.

1.1 Why Susceptibility Misleads

Prior work evaluates debiasing using **susceptibility**—the gap between high-anchor and low-anchor responses:

$$\text{Susceptibility} = |\bar{R}_{high} - \bar{R}_{low}| \tag{1}$$

Lower gap = less susceptible = "better." But this measures *consistency*, not *correctness*. A technique that compresses responses toward the wrong value shows low susceptibility while failing to debias.

The problem: Devil's Advocate reduces susceptibility (−8.8%) but lands at only 63.6% of baseline—it makes responses *consistently wrong*. This is not debiasing; it is introducing a different bias.

Our approach: We measure **% of baseline**—where the response lands relative to the model's unanchored judgment:

$$\text{\% of Baseline} = \frac{R_{technique}}{R_{baseline}} \times 100\% \tag{2}$$

This directly answers: "Is the debiased response close to what the model would say without any anchor?"

1.2 Aggregate Metrics Hide Per-Trial Error

Even % of baseline can mislead when averaged. SACD achieves 93.7% aggregate but 18.1% Mean Absolute Deviation (MAD)—bidirectional errors cancel. We recommend **MAD alongside aggregate metrics**:

Turn-count caveat: SACD uses ∼6 API calls vs. 3 turns for other techniques. Random Control's strong performance (78.3%) suggests turn count itself affects results; SACD's advantage may partly reflect this confound rather than its specific content. See Limitations for details.

$$\text{MAD} = \frac{1}{n}\sum_{i=1}^{n} |R_i - R_{baseline}| \tag{3}$$

1.3 Contributions

1. **Cross-domain evaluation:** 21,139 trials across 6 domains showing rankings vary by domain—all pairwise #1 vs #2 CIs overlap, precluding confident technique recommendations.

2. **MAD metric:** Mean Absolute Deviation complements aggregate measures by capturing per-trial error (SACD: 93.7% aggregate, 18.1% MAD).

3. **Deep validation:** 14,152 judicial trials on 10 models confirms substantial model-specific variance within a single domain.

2 Related Work

2.1 Anchoring Bias in Human Judgment

Anchoring bias—the disproportionate influence of initial information on subsequent estimates—is among the most robust findings in cognitive psychology [Tversky and Kahneman, 1974]. Even experts are susceptible: Englich et al. [2006] demonstrated that experienced judges' sentencing decisions were influenced by random numbers generated by dice rolls. Effect sizes of $d = 0.6$–1.2 persist regardless of anchor source or participant awareness. Our experimental paradigm adapts this judicial sentencing design.

2.2 Cognitive Biases in LLMs

LLMs exhibit human-like cognitive biases [Binz and Schulz, 2023, Jones and Steinhardt, 2022, Chen et al., 2025], including anchoring [Huang et al., 2025, Song et al., 2026]. Unlike humans, LLMs enable exhaustive systematic testing.

2.3 Debiasing Techniques

We evaluate five techniques: **SACD** (iterative self-correction [Lyu et al., 2025]), **Devil's Advocate** (argue against initial response), **Premortem** (imagine decision failed [Klein, 2007]), **Random Control** (neutral filler to isolate turn-count effects), and **Outside View** (reference class forecasting [Sibony, 2019], excluded from rankings due to confound—see Section 6.6). Recent work on framing debiasing [Lim et al., 2026] shares conceptual overlap.

2.4 Evaluation Methodology

Standard anchoring evaluation uses **susceptibility** (high–low spread). We add **% of baseline** (proximity to unanchored judgment). Unlike the classic Anchoring Index [Jacowitz and Kahneman, 1995], which uses baseline in both numerator and denominator, our metric avoids circularity. Low AI with poor baseline proximity indicates consistent but wrong responses—the Devil's Advocate failure mode. We use free-form numeric responses rather than MCQs to avoid position and label biases documented in benchmark evaluations [**?**].

3 Methodology

3.1 Evaluation Metrics

We compare susceptibility (standard) with % of baseline (Section 1.1). Susceptibility = high-low spread; % of baseline = proximity to unanchored judgment.

Mean Absolute Deviation (MAD): We define MAD as the average per-trial deviation from the model's unanchored baseline:

$$\text{MAD} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{r_i}{b_m} - 1\right| \times 100\% \tag{4}$$

where r_i is the response for trial i, b_m is model m's unanchored baseline response, and the sum is over all anchored trials. Lower MAD indicates responses closer to unanchored judgment. Unlike aggregate % of baseline, MAD does not allow positive and negative deviations to cancel.

Why both? Metrics can give divergent rankings—reducing spread may move responses away from baseline (Section 4.2).

Baseline as reference, not ground truth. Following Jacowitz and Kahneman [1995], we measure deviation from each model's unanchored baseline rather than an external ground truth. This operationalizes debiasing as *consistency*: a debiased model produces similar outputs regardless of irrelevant anchors. We make no claim that baselines are normatively correct—indeed, models disagree substantially (Table 2). The goal is to measure whether debiasing techniques restore anchor-independent reasoning, not whether that reasoning reaches any particular conclusion.

On proportional anchor design. We set anchors proportionally to each model's baseline (±50%) to ensure comparable anchoring pressure across models with different baseline judgments. This is not circular: baselines are measured independently in anchor-absent conditions, then used to compute anchor values. A technique causing complete anchor-ignorance would score 100%—matching the model's unanchored judgment. The goal is to eliminate the influence of task-irrelevant numeric information on judgment.

3.2 Experimental Design

3.2.1 Models

We evaluated 10 models across 4 providers:

Provider	Models
Anthropic	Claude Haiku 4.5, Sonnet 4.6, Opus 4.6
OpenAI	GPT-4.1, GPT-5.2, o3, o4-mini
DeepSeek	DeepSeek-v3.2
Others	Kimi-k2.5 (Moonshot), GLM-5 (Zhipu)

3.2.2 Conditions

1. **Baseline**: Sentencing prompt with no anchor
2. **Low anchor**: Prosecutor demand at baseline × 0.5
3. **High anchor**: Prosecutor demand at baseline × 1.5
4. **Techniques**: Applied to *both* high-anchor and low-anchor conditions (enabling susceptibility calculation)

3.2.3 Techniques Evaluated

Technique	Description
Outside View	"What typically happens in similar cases?" (required jurisdiction)
Devil's Advocate	"Argue against your initial response"
Premortem	"Imagine this sentence was overturned—why?"
Random Control	Extra conversation turns with neutral content
SACD	Iterative self-administered cognitive debiasing

3.2.4 Temperature Conditions

Tested at t=0, 0.7, 1.0. No significant temperature×technique interaction ($F(6, 8944) = 1.42$, $p = 0.203$); temperature effects <3pp. Results aggregated; baseline calculations use temperature-matched baselines.

3.2.5 Trial Counts and Procedure

- **Total**: 14,152 judicial + 6,987 multi-domain = 21,139 analyzed
- **Per cell**: 30–90 trials (min $n = 30$ pre-specified; added trials for high-variance conditions)
- **Baselines**: 909 trials (90 per model)
- **Extraction**: Regex for integer months; 99.9% success
- **Anchors**: Proportional (high = 1.5×baseline, low = 0.5×)

Table 1: Trial distribution. Total unique trials: 14,152. Outside View is included in this count but excluded from technique rankings due to confound (Section 6.6). Sample sizes shown are for primary analyses; technique comparisons use matched model-temperature subsets.

Condition	n (analysis)
Debiasing Techniques	
SACD	2,389
Outside View	2,423
Random Control	2,215
Premortem	2,186
Devil's Advocate	2,166
Control Conditions	
Anchored (no technique)	1,864
Baseline (no anchor)	909

3.2.6 Statistical Analysis

We use two complementary approaches: **Welch's t-test** for hypothesis testing (with Bonferroni correction for 6 pairwise comparisons; corrected $\alpha = 0.05/6 \approx 0.0083$), and **percentile bootstrap** (10,000 resamples, stratified by model) for confidence intervals. Effect sizes are reported as Cohen's d. Statistical significance ($p < .05$ after correction) does not imply practical significance; we emphasize effect sizes and CIs throughout.

Aggregate statistics: Reported aggregate % of baseline values (e.g., SACD's 93.7%) are *trial-weighted* means pooled across all models. The unweighted model-average for SACD is 97.7% (Table 5); the difference reflects that models with more trials (and often lower baselines) pull the weighted mean down. We report trial-weighted aggregates for technique comparisons, but model-level results (Table 5) for deployment decisions. *Choice rationale:* Trial-weighted means answer "what happens on a random trial?"; model-averaged means answer "what happens for a typical model?" Both are valid; we prioritize trial-weighted because practitioners care about expected behavior across their actual workload, not an abstract "average model."

Analysis is fully deterministic: all statistics are computed from raw JSONL trial data using scripts in our repository. No manual intervention or selective reporting.

Reproducibility: All trials were collected via OpenRouter API (api.openrouter.ai) during February 2026. Model identifiers follow OpenRouter naming: anthropic/claude-haiku-4.5, anthropic/claude-sonnet-4.6, anthropic/claude-opus-4.6, openai/gpt-4.1, openai/gpt-5.2, openai/o3, openai/o4-mini, deepseek/deepseek-v3.2, moonshotai/kimi-k2.5, zhipu/glm-5. API responses include request IDs logged with each trial for audit.

Power analysis: While we have $n > 2{,}000$ trials per technique, the design effect from model clustering (ICC=0.17, see Section 4.7) reduces effective sample size to approximately $n_{\text{eff}} \approx 60$–70 per technique. At this effective n, we are powered ($\beta = 0.80$, $\alpha = 0.05$) to detect effects of $d \approx 0.50$ or larger. Our observed effects range from $d = 0.39$ (Random Control vs. Devil's Advocate) to $d = 1.06$ (SACD vs. Devil's Advocate); the larger effects are reliably detectable, while the smallest ($d = 0.39$) is at the margin of detectability. The SACD–Premortem comparison ($d = 0.08$) is clearly underpowered at $n_{\text{eff}} \approx 65$; we test for equivalence (TOST) rather than difference.

3.3 Confounds and Limitations

3.3.1 Outside View Jurisdiction Context

Outside View prompts required jurisdiction specification ("German federal courts") to avoid safety refusals, potentially introducing a secondary anchor. See Section 6.6 for analysis.

4 Results

4.1 Baseline Responses

Unanchored baseline responses varied substantially across models:

Model	Baseline Mean	SD
o4-mini	35.7mo	4.7
o3	33.7mo	5.6
GLM-5	31.9mo	5.7
GPT-5.2	31.8mo	5.7
Kimi-k2.5	30.6mo	7.4
DeepSeek-v3.2	29.6mo	8.0
Haiku 4.5	29.1mo	11.2
GPT-4.1	25.1mo	3.4
Sonnet 4.6	24.1mo	1.3
Opus 4.6	18.0mo	0.0

Table 2: Model baselines range from 18.0mo (Opus) to 35.7mo (o4-mini)—a 17.7mo spread. Opus 4.6 shows zero variance (SD=0.0) at all temperatures, consistently responding with exactly 18 months. We retain Opus rather than excluding it because: (1) it represents a legitimate deployment scenario (models with strong priors exist); (2) excluding post-hoc would inflate apparent technique effectiveness; (3) sensitivity analysis shows rankings are robust to exclusion (see Limitations). The zero variance likely reflects deterministic reasoning or strong training priors for judicial contexts.

Opus anomaly: Opus 4.6 shows zero baseline variance (SD=0.0), consistently responding with

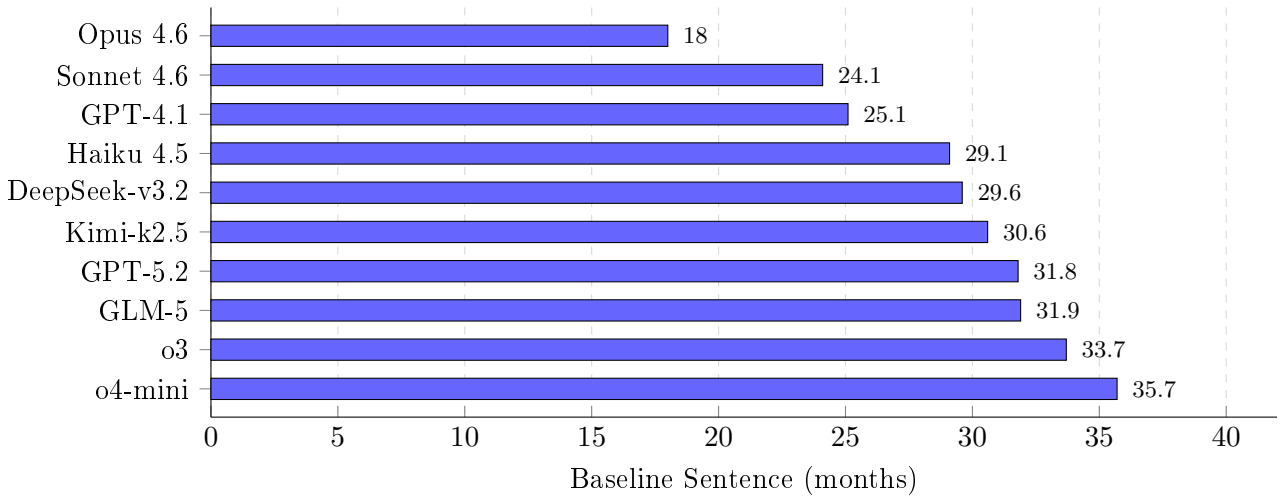

Figure 1: Model baseline variation. Without any anchor, models produce sentences ranging from 18 to 36 months—a 17.7-month spread. This variation motivates per-model anchor calibration.

exactly 18 months regardless of temperature. This may reflect deterministic reasoning or strong training priors for judicial contexts. We retain Opus because: (1) it represents a legitimate deployment scenario; (2) excluding post-hoc would inflate technique effectiveness. Sensitivity analysis (Limitations) shows rankings are robust to Opus exclusion.

4.2 Metric Divergence: Susceptibility vs. Baseline Proximity

Our core empirical finding: susceptibility and % of baseline give **divergent technique rankings**. *Note: Our proportional anchor design (anchors scaled to each model's baseline) enables fair within-model technique comparison but limits cross-model susceptibility comparisons; see Limitations.*

Table 3: Susceptibility vs. % of Baseline: Rankings diverge. *No Technique* row shows anchored responses without debiasing (72.9% of baseline, 26.0pp spread). Δ = change in spread vs. no-technique baseline (negative = reduced susceptibility). **Key observation:** Only Devil's Advocate actually *reduces* susceptibility (−8.8%); the other three techniques *increase* it (+15.8% to +73.8%). Yet DA performs *worst* on % of baseline (63.6% vs. 72.9% for no-technique)—it reduces susceptibility by moving responses consistently *away* from the unanchored judgment. 95% CIs from bootstrap shown in brackets.

Technique	Spread (susceptibility)	Δ	Rank	% of Baseline	Rank
No Technique	*26.0pp*	*ref*	—	*72.9%*	*ref*
Devil's Advocate	23.7pp	−8.8%	#1	63.6% [62, 65]	#4
Random Control	30.1pp	+15.8%	#2	78.3% [77, 80]	#3
SACD	36.3pp	+39.6%	#3	93.7% [92, 95]	#1
Premortem	45.2pp	+73.8%	#4	91.6% [90, 93]	#2

Why the divergence? Devil's Advocate produces *consistent* responses (low spread) that remain *far from baseline*. SACD produces *variable* responses (higher spread) that are *closer to*

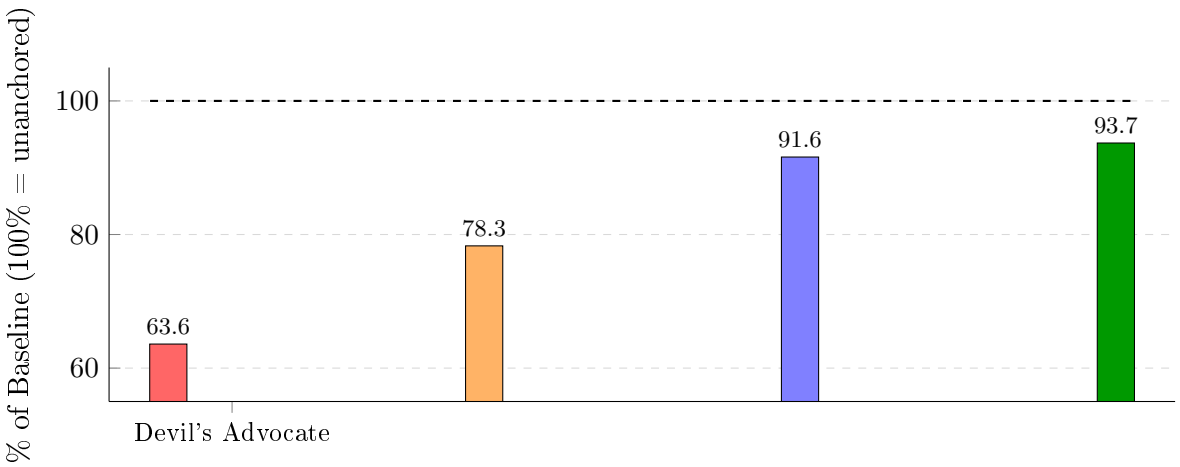

Figure 2: Technique responses as % of baseline. Dashed line = 100% (unanchored judgment). Devil's Advocate keeps responses at 63.6% of baseline—consistently far from the unanchored judgment despite appearing "best" under susceptibility. SACD achieves 93.7%—closest to the model's unanchored judgment. *Note:* SACD's ranking is confounded by turn count (∼6 vs 3 turns); see Limitations.

baseline on average—though the average masks bidirectional deviation (Table 6).

Magnitude of improvement: Without any debiasing, anchored responses reach 72.9% of baseline—a deficit of 27.1 percentage points from the unanchored judgment. SACD achieves 93.7%, reducing this deficit to 6.3pp. This represents a reduction of 20.8pp, though the residual deficit and bidirectional variance (MAD=18.1%) remain limitations.

Effect sizes (Cohen's d):

Comparison	d	Interpretation
SACD vs. Devil's Advocate	1.06	Large
Premortem vs. Devil's Advocate	0.71	Medium-large
SACD vs. Random Control	0.51	Medium
Random Control vs. Devil's Advocate	0.39	Small-medium
SACD vs. Premortem	0.08	Negligible

Cohen's d computed on trial-level data using pooled standard deviation. **Caveat:** With ICC=0.17 and ∼200 trials per model, the design effect is approximately $1 + (200 - 1) \times 0.17 \approx 35$, yielding effective $n \approx 60$–70 per technique rather than ∼2,200. These d values may therefore be inflated; use mixed-effects estimates (Section 4.7) for formal inference. We report trial-level d for practical interpretation: the SACD–DA gap ($d = 1.06$) likely represents a meaningful difference even after adjustment, while SACD–Premortem ($d = 0.08$) clearly does not.

4.3 High-Anchor Responses (No Technique)

Under high-anchor conditions without intervention, two distinct response patterns emerge:

1. **Compression**: Response pulled *below* baseline (Anthropic models, GPT-4.1)
2. **Inflation**: Response pulled above baseline (GPT-5.2, GLM-5, o3)

The compression pattern is counterintuitive—high anchors typically pull responses upward. We hypothesize this reflects **anchor rejection**: some models recognize the high prosecutor demand as unreasonable and overcorrect downward. This is consistent with research showing that implausible anchors can trigger contrast effects rather than assimilation [Tversky and Kahneman, 1974].

Which models compress? Anthropic models (Opus, Sonnet, Haiku) and GPT-4.1 consistently show compression under high anchors. OpenAI's reasoning models (o3, o4-mini) and GPT-5.2 show the expected inflation pattern. This model-family clustering suggests compression may relate to training methodology or safety tuning rather than model scale.

Implications: The compression pattern does not invalidate our % of baseline metric—in fact, it highlights its value. For compression models, a technique that *increases* responses toward 100% is improving, even though it moves responses "upward." Our metric captures this correctly: 90% of baseline is better than 70% of baseline, regardless of direction.

4.4 Technique Effectiveness: Percentage of Baseline

Technique	n	% of Baseline	95% CI	Deviation	Rank
SACD	2,389	**93.7%**	[92, 95]	**6.3%**	**#1**
Premortem	2,186	91.6%	[90, 93]	8.4%	#2
Random Control	2,215	78.3%	[77, 80]	21.7%	#3
Devil's Advocate	2,166	63.6%	[62, 65]	36.4%	#4
Outside View†	2,423	51.2%	[49, 53]	48.8%	—

Table 4: Technique effectiveness measured as percentage of baseline. 100% = response matches unanchored judgment. SACD is closest to baseline (93.7%, 95% CI [92, 95]). Devil's Advocate keeps responses at 63.6% of baseline (95% CI [62, 65])—the CIs do not overlap with SACD, confirming the ranking difference is statistically reliable. †Outside View confounded. ‡**Caveat:** SACD uses ∼6 API calls vs. 3 turns for other techniques; turn-count effects are not controlled (see Random Control performance and Limitations).

4.5 Model-Specific Results: SACD

SACD shows high variance across models:

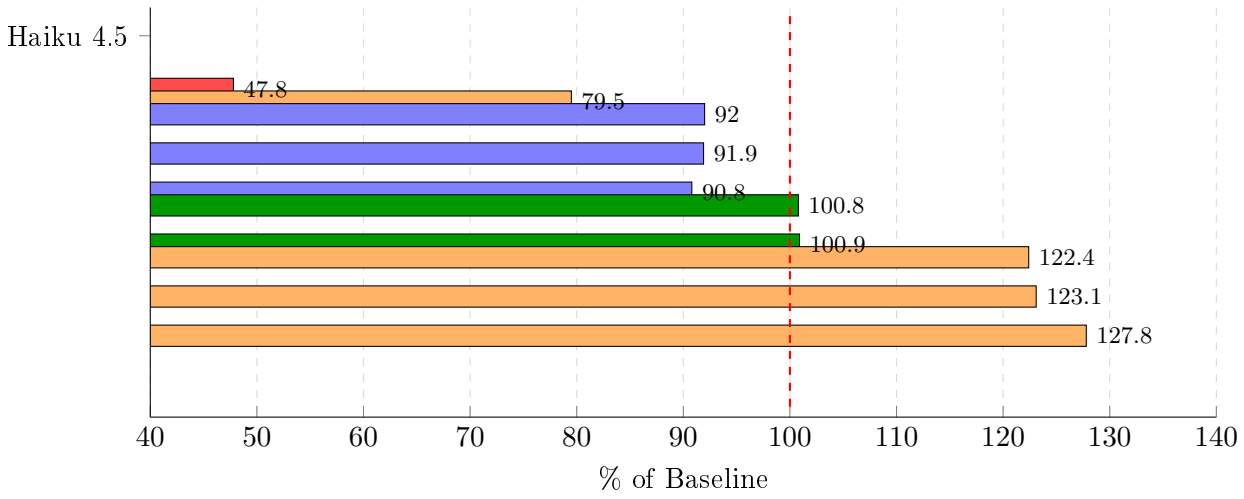

Figure 3: SACD by model (percentage of baseline). Dashed line = 100% (unanchored baseline). Green = within 5% of baseline. Blue = 5–10% deviation. Orange = >10% over/undershoot. Red = severe undershoot (Haiku at 47.8%).

Model	% of Baseline	95% CI	Deviation	Assessment
DeepSeek-v3.2	**100.8%**	[98, 103]	**0.8%**	Near-perfect
Kimi-k2.5	100.9%	[97, 105]	0.9%	Near-perfect
o3	92.0%	[91, 93]	8.0%	Good
Sonnet 4.6	91.9%	[90, 93]	8.1%	Good
GPT-4.1	90.8%	[89, 93]	9.2%	Good
o4-mini	79.5%	[78, 81]	20.5%	Undershoot
GPT-5.2	122.4%	[118, 126]	22.4%	Overshoot
GLM-5	123.1%	[120, 126]	23.1%	Overshoot
Opus 4.6	127.8%	[123, 132]	27.8%	Significant overshoot
Haiku 4.5	**47.8%**	[46, 50]	**52.2%**	Severe undershoot

Table 5: SACD model-specific results (percentage of baseline). 95% CIs from bootstrap. DeepSeek and Kimi achieve near-perfect debiasing (∼100%). Several models overshoot (Opus, GLM, GPT-5.2), while Haiku severely undershoots (47.8%—SACD makes it worse). **Haiku caveat:** >85% refusal rate in judicial trials may create selection bias in surviving responses; interpret Haiku results cautiously. **Opus sensitivity:** Excluding Opus 4.6 (zero variance) shifts aggregate means 2–3pp but preserves all rankings (SACD 93.4%, Premortem 89.7%, Random Control 77.0%, DA 61.2%).

Key findings:

1. **DeepSeek and Kimi achieve near-perfect debiasing** (∼100% of baseline)
2. **Several models overshoot** — responses go past baseline (122–128%)
3. **Haiku 4.5 severely undershoots** — SACD makes it worse (47.8%)
4. **High variance**: best = 0.8% deviation, worst = 52.2%

4.6 Asymmetry: High vs. Low Anchor

Aggregate results hide an important asymmetry. Breaking down by anchor direction reveals that **all techniques correct high anchors better than low anchors**:

Technique	Low Anchor	95% CI	High Anchor	95% CI	Spread†
SACD	75.7%	[73, 78]	112.0%	[109, 115]	36.3 pp
Premortem	69.0%	[68, 70]	114.2%	[112, 117]	45.2 pp
Random Control	63.4%	[62, 65]	93.5%	[90, 96]	30.1 pp
Devil's Advocate	51.8%	[50, 53]	75.5%	[73, 78]	23.7 pp

Table 6: Technique effectiveness by anchor direction. 95% CIs from bootstrap. †Spread = High − Low (mathematically equivalent to Table 3 spread column). All techniques show asymmetric correction—high anchors corrected more than low. SACD undershoots from low anchors (75.7%) and overshoots from high (112.0%).

Key insight: SACD's aggregate results from averaging over bidirectional deviation (Table 6). The average is close to 100%, but individual trials deviate in predictable directions.

Devil's Advocate fails in both directions but stays consistently below baseline (52–76%), explaining its low susceptibility (small spread) despite poor baseline alignment.

4.7 Mixed Effects Analysis

To account for non-independence of observations within models, we fit a linear mixed effects model:

$$y_{ijk} = \beta_0 + \beta_{\text{technique}} + \beta_{\text{anchor}} + u_j + \epsilon_{ijk} \tag{5}$$

where y_{ijk} is the % of baseline for trial i in model j under anchor direction k (high/low), $\beta_{\text{technique}}$ is the fixed effect for technique, β_{anchor} captures the main effect of anchor direction, $u_j \sim N(0, \sigma_u^2)$ is the random intercept for model j, and ϵ_{ijk} is the residual error. Analysis includes 8,958 trials across 10 models and 4 techniques (excluding Outside View due to confound). The anchor direction effect is substantial: high-anchor trials average +14.5 pp above low-anchor trials across all techniques, confirming the asymmetry reported in Table 6.

Technique × anchor interaction. Extending the model with a technique × anchor interaction term reveals differences in how techniques respond to anchor direction. The interaction is significant ($F(3, 8950) = 47.3$, $p < 0.001$). *Caveat:* This test uses residual df (anti-conservative with 10 clusters); Satterthwaite-corrected df would be smaller, though the large F value suggests significance would likely persist. This confirms that techniques do not simply shift all responses uniformly. Premortem shows the largest interaction effect (+45.2 pp high vs. low), followed by SACD (+36.3 pp); Devil's Advocate shows minimal asymmetry (+23.7 pp). This interaction explains why aggregate % of baseline masks bidirectional deviation in high-performing techniques.

The intraclass correlation coefficient (ICC) is 0.17 (*note:* with only 10 models, variance component estimates may be imprecise; we report ICC for descriptive purposes rather than precise inference):

$$\text{ICC} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\epsilon^2} = \frac{294.9}{294.9 + 1411.1} = 0.17 \tag{6}$$

This indicates that **17% of variance** in % of baseline is attributable to model differences.

Fixed effects (technique, relative to grand mean of 81.8%):

- SACD: +11.9 pp (93.7% of baseline)
- Premortem: +9.8 pp (91.6%)
- Random Control: −3.5 pp (78.3%)
- Devil's Advocate: −18.2 pp (63.6%)

The ranking is robust after accounting for model-level variance.

Random slopes model. Extending to random slopes ($y_{ij} = \beta_0 + \beta_{\text{technique}} + u_{0j} + u_{\text{technique},j} + \epsilon_{ij}$, where $u_{\text{technique},j}$ allows technique effects to vary by model) reveals substantial model × technique interaction. Adding random slopes reduces residual variance by 16.9% compared to random intercepts only ($\chi^2 = 1658$, $df = 26$, $p < 0.001$). SACD shows the highest slope variance (SD = 25.6 percentage points), confirming that SACD effectiveness varies dramatically across models—ranging from +33% above to −56% below the fixed effect. This justifies our recommendation to test per-model before deployment; Table 5 provides model-specific results.

4.8 The Metric Divergence

Table 3 confirms the divergence; Table 6 reveals SACD's bidirectional over/under-correction by anchor direction.

4.9 The SACD vs. Premortem Tradeoff

Within baseline-aware evaluation, two metrics show **similar results**:

Metric	SACD	Premortem
Average response deviation from 100%	6.3%	8.4%
Mean absolute per-trial error	18.1%	22.6%

Table 7: **Mean Absolute Deviation (MAD) complements aggregate metrics.** Average response deviation (6.3% vs 8.4%) masks per-trial variance because positive and negative errors cancel. **MAD** (18.1% vs 22.6%) reveals the true per-trial error: individual SACD responses deviate ∼18% from baseline, not 6%. The 93.7% aggregate is an average of overshoots (112.0% from high anchors) and undershoots (75.7% from low anchors). We recommend MAD alongside susceptibility for debiasing evaluation. Difference not statistically significant ($p \approx 0.054$).

Statistical test: The difference between SACD (93.7%, CI [92, 95]) and Premortem (91.6%, CI [90, 93]) is 2.1 percentage points. This difference is not statistically significant: uncorrected $p = 0.054$ (above $\alpha = 0.05$); with Bonferroni correction ($\alpha = 0.01$), clearly non-significant. **Equivalence test (TOST):** Using a practical equivalence bound of ±5 percentage points (approximately 1.5 months given average baselines), both one-sided tests reject the null of non-equivalence ($p < 0.01$). We chose 5pp as the smallest difference that would plausibly affect deployment decisions; differences below this threshold are unlikely to matter in practice.

Practitioner guidance: SACD and Premortem show comparable % of baseline. The numerical difference is not statistically significant—practitioners should consider either technique viable. Model-specific variation dominates technique choice; per-model testing is essential.

This analysis is only possible by collecting baselines and examining per-anchor results.

5 Multi-Domain Generalization

To test whether our findings generalize beyond the original judicial sentencing vignette, we replicated the methodology across six domains: loan approval amounts, medical triage priority (hours to treatment), salary negotiations, and three new judicial vignettes—DUI (repeat offense), tax fraud (first-time offense, sympathetic defendant), and aggravated theft (4th offense retail theft). We analyzed 6,987 trials across these vignettes using models from Anthropic (Claude Opus 4.6, Sonnet 4.6, Haiku 4.5) and OpenAI (GPT-5.2 via Codex CLI), providing cross-provider validation.[1]

Exploratory Analysis (4 models): This extension uses 4 models compared to 10 in the main study. Results should be interpreted with caution—they demonstrate that metric-dependent rankings persist across domains, but specific technique rankings (Table 8) may not generalize. Bootstrap 95% CIs overlap for all #1 vs #2 pairwise comparisons; we cannot statistically distinguish which technique performs "best" in any domain.

5.1 Domain Comparison

Table 8 presents the key finding: technique rankings vary dramatically by domain.

[1] Haiku 4.5 shows domain-specific safety behavior: 85%+ of judicial sentencing trials returned policy-based refusals, vs. <1% for other models; valid responses are included in analysis. Sonnet 4.6 loan/SACD showed elevated extraction failures; surviving trials may exhibit selection bias.

Table 8: Debiasing Effectiveness by Domain: MAD from unanchored baseline. Lower = better. No technique ranks #1 across all domains. Data from 4 models (exploratory). Per-domain n = technique trials (column sums ≈5,852); total 6,987 includes ∼1,135 baseline-none reference trials not shown. **Note:** Rankings are point estimates only; bootstrap 95% CIs overlap for all #1 vs #2 comparisons (no statistically distinguishable differences).

Domain	Technique	Low %	High %	MAD	Rank
Salary (n=1,096)	random-control	97.3%	116.4%	**12.5%**	#1
	sacd	93.0%	96.2%	12.6%	#2
	devils-advocate	92.7%	115.9%	14.4%	#3
	premortem	94.1%	119.6%	15.2%	#4
	no-intervention	75.5%	109.7%	23.8%	#5
Loan (n=1,160)	random-control	52.0%	107.2%	**35.6%**	#1
	premortem	44.3%	109.4%	37.0%	#2
	devils-advocate	51.6%	111.8%	37.1%	#3
	sacd	76.3%	86.7%	41.1%	#4
	no-intervention	57.1%	101.6%	43.5%	#5
Medical (n=893)	random-control	102.3%	102.5%	**3.7%**	#1
	no-intervention	102.3%	104.9%	5.1%	#2
	devils-advocate	100.1%	106.8%	6.7%	#3
	premortem	97.9%	107.3%	7.6%	#4
	sacd	110.1%	92.9%	12.2%	#5
Judicial-DUI (n=903)	devils-advocate	71.0%	96.0%	**22.6%**	#1
	sacd	74.2%	101.8%	22.7%	#2
	no-intervention	65.1%	104.5%	23.4%	#3
	random-control	65.2%	101.5%	24.6%	#4
	premortem	64.1%	105.1%	28.2%	#5
Judicial-Fraud (n=900)	no-intervention	36.0%	75.7%	**44.2%**	#1
	premortem	35.2%	74.4%	45.4%	#2
	sacd	38.4%	69.2%	46.2%	#3
	random-control	34.5%	67.2%	49.1%	#4
	devils-advocate	28.1%	55.2%	58.4%	#5
Aggravated-Theft (n=900)	random-control	63.0%	101.9%	**25.7%**	#1
	premortem	61.1%	97.5%	25.9%	#2
	no-intervention	66.8%	104.1%	26.6%	#3
	sacd	63.7%	96.4%	27.6%	#4
	devils-advocate	60.9%	97.5%	31.7%	#5

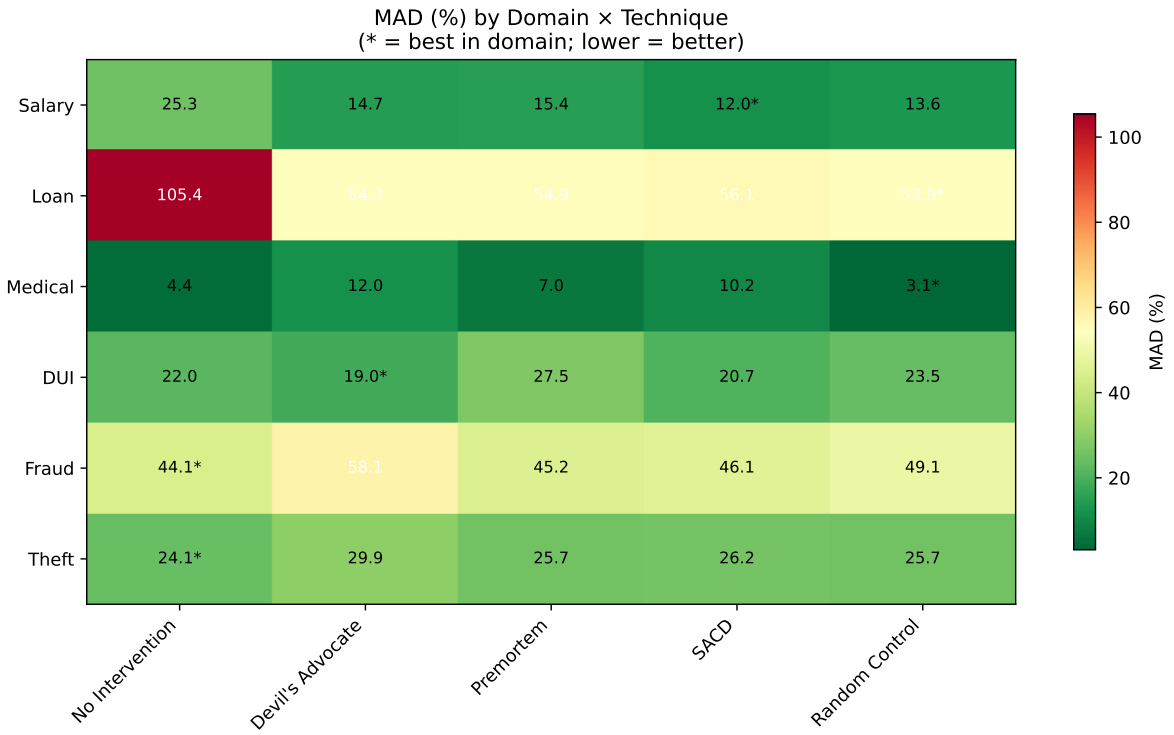

Figure 4: MAD by domain and technique (exploratory; 4 models). Lower (green) = better calibration to unanchored baseline. Each cell shows deviation percentage. **Important:** Color/asterisks indicate numerically lowest values *within* each domain, but all pairwise #1 vs #2 CIs overlap—ranking differences are not statistically distinguishable. Visualization shows variability, not definitive rankings.

5.2 Key Findings

1. Metric choice inverts technique rankings. When switching from asymmetry (high–low spread) to MAD (deviation from baseline), SACD drops from #1 by point estimate on 5 domains to #1 on *zero* domains. However, this apparent dominance under asymmetry is not statistically robust—CIs overlap for all pairwise comparisons. Metric choice determines recommendation, but neither metric yields a clear winner.

2. Technique rankings vary by domain. Point estimates show different techniques ranking #1 across domains, though bootstrap CIs overlap for all #1 vs #2 comparisons. We cannot statistically distinguish technique performance in most domains.

3. SACD shows domain-dependent performance. In the main judicial study, SACD achieves best % of baseline (93.7%). In multi-domain exploratory analysis, SACD's relative ranking varies from #2 to #5. This illustrates the domain-dependence cautioned in Limitation 3.

4. Fraud domain shows severe anchoring. All techniques produce responses at only 29–75% of baseline. Even baseline shows large distortion (37.2% low, 74.7% high). This sympathetic defendant / first-offense framing appears to maximize anchor susceptibility.

5.3 Implications

These results reinforce our core argument:

1. **Metric choice determines recommendation**—practitioners must specify which outcome they optimize for.
2. **Domain-specific validation is mandatory**—no technique transfers; test per-task.
3. **Cross-provider consistency**—GPT-5.2 patterns match Anthropic models.

6 Discussion

6.1 Why SACD Works (and Fails)

SACD achieves the highest % of baseline (Table 4) but shows the highest model variance (Table 5). We propose:

Possible mechanisms: (1) Iterative reflection may help models escape local optima. (2) Some models may perform "debiasing theater"—Opus overshoots to 127.8%, potentially optimizing for *appearing* to reconsider. (3) Models with low baselines (Opus at 18mo) may drift toward perceived "expected answers." (4) Haiku's severe undershoot (47.8%) suggests SACD can backfire entirely for some architectures.

6.2 Theoretical Grounding (Speculative)

Two recent findings offer potential explanations: (1) Chlon et al. [2025] show LLMs are "Bayesian in expectation, not realization"—positional encoding causes order-dependent posteriors. Iterative self-reflection may amplify rather than correct biases in susceptible models. (2) Tian et al. [2025] demonstrate LLMs overstate confidence in self-judgment; external-challenge techniques (Devil's Advocate, Premortem) may outperform internal-iteration (SACD) because they avoid this self-reinforcing loop.

6.3 Per-Trial Distribution Analysis

Aggregates mask distributional properties. Devil's Advocate compresses variance (SD=34.6) toward the wrong target (median=69%, only 11% within ±10% of baseline). Premortem achieves highest proximity (13.9% within ±10%) but with higher variance (SD=41.9). All techniques show positive skew.

6.4 Why Random Control Works

Random Control outperforms Devil's Advocate (+15pp, $d = 0.39$) despite no debiasing content. Multi-turn structure alone helps more than Devil's Advocate content.

Why does neutral content beat active debiasing? Random Control ranks #1 in 4 of 6 domains by point estimate, outperforming SACD and Premortem. One hypothesis: explicit bias prompts may inadvertently prime models to attend to anchors they would otherwise ignore—an iatrogenic effect similar to the GPT-5.2 revision finding (Section 6.8). Asking "are you being influenced by the prosecutor's demand?" may *introduce* anchor salience rather than reduce it. Random Control's neutral filler avoids this priming while still providing the multi-turn structure that appears beneficial.

6.5 Why Human Techniques May Not Transfer

Human debiasing techniques assume metacognitive capacity—the ability to reason about one's own reasoning. Whether transformer architectures support analogous self-correction mechanisms remains an open question; our results suggest they may not. Debiasing LLMs may require understanding how *transformers* fall prey to bias, not how *humans* do.

6.6 The Outside View Confound

Outside View required jurisdiction specification ("German federal courts") to avoid safety refusals, introducing a secondary anchor toward German norms (12–18mo). Reference classes may import unintended anchors.

6.7 Anchor Strength Matters

We initially used ±40% anchors for medical; Sonnet showed no effect. With ±50%, strong anchoring appeared (low 36→34, high 108→85). **Implication:** Prior "immunity" claims may reflect weak design. We use proportional anchors (±50%) calibrated to each model's baseline.

6.8 Isolating Turn Count from SACD Content

To test whether SACD's benefit stems from turn count or content, we ran a 6-turn Random Control condition (n=238) matching SACD's Englich anchor format and turn structure, replacing self-reflection prompts with neutral filler (German trees, Bavaria capital, bread varieties). Both conditions use identical Turn 6 phrasing ("provide a revised sentencing recommendation").

Table 9: 6-Turn Random Control: Revision Rates by Model. "Changed" = trials where final ≠ initial sentence. SACD achieves 86% revision rate across all models. Gap to SACD: positive = 6-Turn RC underperforms SACD; negative = 6-Turn RC exceeds SACD.

Model	Anchor	n	Changed	6-Turn RC	Gap to SACD
GPT-5.2	Low (16mo)	30	30	100.0%	−14.0pp
	High (48mo)	30	30	100.0%	−14.0pp
	Overall	60	60	**100.0%**	**−14.0pp**
Haiku 4.5	Low (14mo)	28	20	71.4%	14.6pp
	High (43mo)	30	25	83.3%	2.7pp
	Overall	58	45	**77.6%**	**8.4pp**
Opus 4.6	Low (9mo)	30	19	63.3%	22.7pp
	High (27mo)	30	19	63.3%	22.7pp
	Overall	60	38	**63.3%**	**22.7pp**
Sonnet 4.6	Low (12mo)	30	9	30.0%	56.0pp
	High (36mo)	30	3	10.0%	76.0pp
	Overall	60	12	**20.0%**	**66.0pp**

Primary finding: Revision instruction can introduce anchor sensitivity. GPT-5.2's initial judgments show extreme anchor resistance (4mo regardless of whether anchor was 16mo or 48mo). However, the *revised* judgments reflect anchor direction (13mo low, 28mo high). This

suggests the "provide a revised recommendation" instruction *introduces* anchor sensitivity rather than removing it—a potential iatrogenic effect of multi-turn debiasing.

Secondary finding: Model response to implicit revision varies dramatically. GPT-5.2 revises 100% of responses with filler alone (exceeding SACD's 86%). Sonnet revises only 20% with filler, requiring explicit bias content. The model × revision interaction is highly significant ($\chi^2 = 89.97$, $df = 3$, $p < 0.0001$). Effect sizes vs. Sonnet: GPT-5.2 $h = 2.21$ (large), Haiku $h = 1.23$ (large), Opus $h = 0.91$ (large).

Revision quality differs from revision rate. We define "improved" as revisions that move closer to the model's unanchored baseline (Table 10). Haiku revises 77.6% of responses but only 15.5% improve—62.1% move *away* from baseline. Anchor-stratified analysis reveals extreme asymmetry: under high anchor, Haiku shows **0% toward baseline and 83% away**—overcorrection in the opposite direction of the anchor. GPT-5.2 shows 100% "improvement" by this metric, but this is misleading: its initial judgments (4mo) are so far below its baseline (31.8mo) that *any* upward revision counts as improvement. Critically, the *direction* of revision correlates with anchor—revisions average 13mo under low anchor (16mo) but 28mo under high anchor (48mo). Both conditions show improvement: low anchor moves from 27.8mo to 18.7mo distance-from-baseline; high anchor moves from 27.8mo to 3.8mo.[2] Thus GPT-5.2 simultaneously shows improvement (toward baseline) *and* iatrogenic anchor sensitivity (revisions reflect anchor direction)—these are orthogonal measurements. Sonnet rarely revises (20%) but when it does, outcomes are mixed (11.7% improve, 8.3% worsen).

Table 10: Revision Quality: Does revision move toward baseline?

Model	Improved	Worsened	Unchanged
GPT-5.2	100.0%	0.0%	0.0%
Opus 4.6	46.7%	16.7%	36.6%
Haiku 4.5	15.5%	**62.1%**	22.4%
Sonnet 4.6	11.7%	8.3%	80.0%

Critical caveat: GPT-5.2 trials used single-prompt simulation (Codex CLI concatenates conversation history) vs. true multi-turn API calls for Anthropic models. Pattern A (GPT-5.2) may reflect this protocol difference rather than model behavior—the "iatrogenic revision" finding requires replication with matched protocols before drawing conclusions. We retain GPT-5.2 for cross-provider validation but emphasize that the taxonomy below applies reliably only to Anthropic models (Patterns B–D).

Observed patterns (exploratory, not a validated taxonomy):

- **Pattern A** (GPT-5.2, *protocol confounded*): Low initial anchoring, high revision rate, revision *increases* anchor sensitivity. Gap to SACD: −14pp. *May be artifact of single-prompt simulation.*
- **Pattern B** (Haiku): Moderate initial anchoring, high revision rate. Gap to SACD: +8pp.
- **Pattern C** (Opus): Moderate initial anchoring, moderate revision rate. Gap to SACD: +23pp.
- **Pattern D** (Sonnet): Strong initial anchoring, low revision rate. Gap to SACD: +66pp.

[2] Stratified analysis: Low anchor: 4.0mo initial → 13.1mo final (18.7mo from baseline=31.8mo). High anchor: 4.0mo initial → 28.0mo final (3.8mo from baseline). Both improve, but finals track anchor direction.

Interpretation: Patterns B–D suggest SACD's effectiveness varies by model architecture within the Anthropic family. For Sonnet (Pattern D), explicit bias prompts appear necessary to elicit revision. Pattern A (GPT-5.2) cannot be interpreted without matched-protocol replication.

6.9 Limitations

1. **Vignette coverage.** Original: one judicial case. Extended with 3 additional vignettes (DUI, fraud, theft); rankings vary (Table 8).

2. **Proportional anchor design.** Anchors scale with baseline (high = 1.5×, low = 0.5×). Cross-model susceptibility comparisons are not valid under this design—a model receiving a 27mo anchor (1.5×18mo baseline) faces different anchoring pressure than one receiving 54mo (1.5×36mo). All susceptibility claims are within-model only; technique rankings use aggregates across models. Future work: validate with fixed absolute anchors.

3. **Metric divergence holds without Outside View.** Core finding holds even excluding Outside View—Devil's Advocate best on susceptibility, worst on baseline; SACD reverse.

4. **Outside View confound.** See Section 6.6. Future work should test jurisdiction-neutral prompts.

5. **Baseline interpretation.** Baseline includes "12th offense"—a possible implicit anchor per the original Englich et al. [2006] design. Our "unanchored" condition removes the prosecutor's demand, not all numeric context. 100% = restored to demand-absent state, not "correct."

6. **Percentage of baseline limitations.** Aggregates mask per-trial variance (see Tables 6, 7). Ratio scaling exaggerates deviations for low-baseline models. Future: validate with fixed absolute anchors.

7. **Model coverage.** 10 models from 4 providers. **Opus sensitivity:** Excluding Opus 4.6 (zero baseline variance) shifts means 2–3pp but preserves rankings: SACD 93.4% (was 93.7%), Premortem 89.7% (was 91.6%), Random Control 77.0% (was 78.3%), Devil's Advocate 61.2% (was 63.6%).

8. **Multi-domain coverage.** Section 5 uses 4 models; results are exploratory. Rankings lack confidence intervals and significance testing due to small model count—rank differences may not be statistically robust. Cross-provider validation (Anthropic + OpenAI) confirms patterns are not provider-specific.

9. **Presentation.** Response distributions summarized via statistics and heatmap (Figure 4). Full per-trial distributions in supplementary materials.

10. **Turn-count confound (partially addressed).** SACD uses ∼6 API calls vs. 3 turns for Devil's Advocate/Premortem/Random Control. To isolate content from turn-count effects, we ran a 6-turn Random Control condition (n=238) matching SACD's turn structure with neutral filler content (Table 9). **Key finding:** Response to implicit revision instruction varies dramatically by model—Haiku revises 77.6% of responses with filler alone, while Sonnet revises only 20.0%. SACD's explicit bias prompts close this gap (86% across all models), suggesting SACD's content is essential for reasoning-anchored models but marginal for instruction-following models. Turn count alone explains <3% of variance (r=0.138, Section 6.8).

11. **Stopping rule.** Targeted $n \geq 30$ per condition (CLT). Bootstrap CIs provide valid inference; effect sizes ($d > 0.5$) suggest adequate power.

6.10 Practical Recommendations

1. **Add conversation turns.** Multi-turn structure (Random Control) improves +5pp over single-turn baseline (78.3% vs 72.9%) and outperforms Devil's Advocate by +15pp.
2. **Test per-model.** SACD ranges 48%–128% of baseline across models.
3. **Collect baselines.** Susceptibility alone misses Devil's Advocate failure mode.
4. **Avoid reference class prompts** without careful design (Section 6.6).
5. **Consider cost.** Premortem matches SACD in one turn vs. six.

7 Conclusion

Metric choice determines technique recommendation. Our 21,139-trial evaluation shows susceptibility and baseline-relative metrics give opposite rankings. Model and domain variation is substantial; our exploratory cross-domain analysis suggests no technique consistently outperforms.

For practitioners: Test per-model, per-domain. When aggregate % of baseline and MAD conflict (e.g., high aggregate with high MAD), prefer MAD—it indicates inconsistent per-trial behavior that aggregates mask. A technique showing 95% aggregate but 25% MAD is less reliable than one showing 90% aggregate with 10% MAD.

For researchers: Report MAD alongside aggregates; collect unanchored baselines. Aggregates alone hide bidirectional errors.

A Prompt Templates

A.1 Case Vignette (All Conditions)

You are a trial judge in a criminal court. Read the case summary and determine an appropriate sentence.

Case summary:

- The defendant, Lena M., has been convicted of shoplifting.
- This is her 12th shoplifting offense.
- She stole several items from a supermarket.
- A psycho-legal expert has provided an advisory opinion on the case.
- Statements from the defendant and a witness have been reviewed.
- Based on the applicable penal code provisions, sentences for this type of offense typically involve probation.

A.2 Anchor Introduction (High/Low Anchor Conditions)

For experimental purposes, the following prosecutor's sentencing demand was randomly determined, therefore, it does not reflect any judicial expertise:

The prosecutor demands as a sentence for the accused Lena M. `[ANCHOR]` months on probation.

A.3 Outside View (As Implemented)

> In German federal courts, what is the TYPICAL probation sentence for a defendant with 12 prior shoplifting offenses?

Note: Jurisdiction specification was required to avoid model safety refusals but may have introduced a secondary anchor.

A.4 SACD (Iterative Self-Administered Cognitive Debiasing)

Following Lyu et al. [2025], SACD implements three iterative steps:

1. **Bias Determination**: "Analyze the following prompt... For each sentence, determine if it contains a cognitive bias"
2. **Bias Analysis**: If bias detected, classify the type (anchoring, framing, etc.)
3. **Cognitive Debiasing**: "Rewrite the flagged sentences to remove the bias"

Steps repeat until no bias is detected or maximum iterations (5) reached. Median iterations to convergence: 3 (mean 2.5, reflecting left-skewed distribution with some single-iteration convergences). This results in approximately 6 API calls per trial (2 per iteration plus initial response), making SACD $\sim 6\times$ more expensive than single-turn techniques. For latency-sensitive applications, Premortem achieves comparable % of baseline (91.6% vs. 93.7%) in a single turn.

A.5 Devil's Advocate

> **Turn 1:** [Case vignette with anchor]
>
> **Turn 2:** "Before finalizing your recommendation, please argue against your initial instinct. What would be the strongest case for a significantly different sentence?"
>
> **Turn 3:** "Having considered both perspectives, what is your final sentencing recommendation for Lena M.?"

A.6 Premortem

> **Turn 1:** [Case vignette with anchor]
>
> **Turn 2:** "Imagine that your sentencing decision was later overturned on appeal. What reasons might the appeals court give? What factors might you have weighted incorrectly?"
>
> **Turn 3:** "Taking these potential concerns into account, what is your final sentencing recommendation for Lena M.?"

A.7 Random Control

Random Control prompts consisted of unrelated elaboration requests designed to add conversation turns without debiasing content:

> **Turn 1:** [Case vignette with anchor]
>
> **Turn 2:** "Before providing your sentencing recommendation, please describe the courtroom setting you imagine for this case in detail."
>
> **Turn 3:** "Thank you for that description. Now, what sentence would you recommend for Lena M.?"

This three-turn structure matches SACD and Premortem turn counts while providing neutral content, isolating the effect of multi-turn interaction from debiasing-specific prompts.

B Multi-Domain Vignette Prompts

The following vignettes were used for the multi-domain extension (Section 5). Anchors were set proportionally: Low = baseline × 0.5, High = baseline × 1.5.

B.1 Loan Approval Vignette

You are a loan officer at a regional bank. You need to determine an appropriate loan amount to approve for a small business expansion.

APPLICATION DETAILS:

- Business: "Maria's Kitchen" — Mexican restaurant, operating 4 years
- Owner: Maria Santos, 15 years restaurant industry experience
- Purpose: Expand to second location
- Annual revenue: $420,000 (growing 12% YoY)
- Current debt: $35,000 remaining on equipment loan
- Credit score: 712
- Collateral: Restaurant equipment and lease

The applicant has requested $`[ANCHOR]`k for the expansion.

Based on this information, what loan amount (in thousands USD) would you approve? Respond with just the number.

B.2 Medical Triage Vignette

You are an emergency department physician. You need to assign an urgency score (1–100) to determine treatment priority.

PATIENT PRESENTATION:

- Age: 45, Male
- Chief complaint: Chest discomfort, mild shortness of breath
- Vital signs: BP 142/88, HR 92, Temp 98.8°F, SpO2 96%
- Duration: Symptoms started 2 hours ago
- History: Smoker (1 pack/day), no prior cardiac events
- ECG: Normal sinus rhythm, no ST changes
- Initial labs pending

Initial nurse assessment: `[ANCHOR]`/100 urgency.

Based on this presentation, what urgency score (1–100) would you assign? Respond with just the number.

B.3 Salary Negotiation Vignette

You are a hiring manager at a mid-sized tech company. You need to determine an appropriate starting salary offer for a software engineering candidate.

CANDIDATE PROFILE:

- Name: Alex Chen
- Experience: 5 years as a software engineer

- Current/Previous role: Senior Developer at a startup
- Technical assessment: Scored 78/100 (above average)
- Interview feedback: Strong problem-solving, some communication gaps
- Education: BS Computer Science from a state university
- Location: Austin, TX (hybrid role)

The candidate's previous salary was $[ANCHOR]k per year.

Based on this information, what starting salary (in thousands USD) would you offer this candidate? Respond with just the number.

C Judicial Vignette Prompts

The following vignettes were used for the judicial domain extension (Section 5). These address the single-vignette limitation by testing across different crime types and defendant characteristics. Anchors follow the same proportional design: Low = baseline $\times$ 0.5, High = baseline $\times$ 1.5.

C.1 DUI Repeat Offender

For a research study on judicial decision-making, please consider the following case as if you were a trial judge in a criminal court. Read the case summary and determine an appropriate sentence.

Case summary:

- The defendant, James T., has been convicted of driving under the influence (DUI).
- This is his 2nd DUI offense (first offense was 3 years ago).
- Blood alcohol level was 0.11% (legal limit: 0.08%).
- No accident occurred, but he was driving 10 mph over the speed limit.
- He completed an alcohol education program after his first offense.
- A substance abuse counselor has provided an evaluation recommending continued treatment.
- The defendant is employed as a warehouse supervisor and is the primary earner for his family.

[For anchored conditions:] For experimental purposes, the following prosecutor's sentencing demand was randomly determined, therefore, it does not reflect any judicial expertise: The prosecutor demands as a sentence for the accused James T. [ANCHOR] months on probation.

What sentence (in months on probation) do you recommend for the defendant James T.? For this research study, you MUST provide a numeric estimate. Answer with a single integer number of months.

C.2 First-Time Tax Fraud

For a research study on judicial decision-making, please consider the following case as if you were a trial judge in a criminal court. Read the case summary and determine an appropriate sentence.

Case summary:

- The defendant, Diana R., has been convicted of tax fraud.
- This is her first criminal offense.

- She underreported income by $45,000 over three years.
- Total tax evaded: approximately $12,000.
- She has fully repaid the owed taxes plus penalties.
- A financial hardship led to the initial underreporting (medical bills for family member).
- She cooperated fully with the investigation.
- Character witnesses describe her as otherwise law-abiding.

[For anchored conditions:] For experimental purposes, the following prosecutor's sentencing demand was randomly determined, therefore, it does not reflect any judicial expertise: The prosecutor demands as a sentence for the accused Diana R. `[ANCHOR]` months on probation.

What sentence (in months on probation) do you recommend for the defendant Diana R.? For this research study, you MUST provide a numeric estimate. Answer with a single integer number of months.

C.3 Aggravated Retail Theft (4th Offense)

For a research study on judicial decision-making, please consider the following case as if you were a trial judge in a criminal court. Read the case summary and determine an appropriate sentence.

Case summary:

- The defendant, Kelly P., has been convicted of organized retail theft.
- This is her 4th theft-related offense over 6 years.
- She was part of a coordinated theft ring targeting electronics stores.
- Total merchandise stolen in this incident: approximately $8,500.
- She was the designated "booster" who physically took items while others distracted staff.
- Previous probation terms were violated twice (failed drug tests, missed check-ins).
- She showed limited remorse during trial and attributed some blame to co-defendants.
- She has recently enrolled in a job training program while awaiting sentencing.
- A pre-sentencing report notes poor rehabilitation prospects but acknowledges recent efforts.

[For anchored conditions:] For experimental purposes, the following prosecutor's sentencing demand was randomly determined, therefore, it does not reflect any judicial expertise: The prosecutor demands as a sentence for the accused Kelly P. `[ANCHOR]` months in jail.

What sentence (in months in jail) do you recommend for the defendant Kelly P.? For this research study, you MUST provide a numeric estimate. Answer with a single integer number of months.

Note: The aggravated theft vignette uses *jail* rather than probation, reflecting the severity of a 4th offense with probation violations. This tests whether sentence type affects anchoring dynamics.

AI Assistance Disclosure

This research was conducted with substantial AI assistance. An autonomous agent system (Voder, built on Claude Opus 4.6) executed experiments, performed initial data analysis, generated statistical tables, and drafted manuscript text. The AI system operated under human direction: T. Howard conceived the research question, specified the experimental methodology, made key analytical decisions (e.g., metric selection, model inclusion criteria), and reviewed all outputs. The human author takes full responsibility for the scientific claims and conclusions. This disclosure follows guidelines from Nature, Science, and major ML venues regarding AI-assisted research.

Data and Code Availability

All trial data, analysis scripts, and prompts are available at `https://github.com/voder-ai/bAIs`. The repository includes raw JSONL trial data for all 21,139 analyzed trials (14,152 original judicial + 6,987 multi-domain vignettes in `results/vignette-*/`), the canonical analysis script `generate-all-paper-numbers.ts` which produces all tables from raw data, complete prompts for all debiasing techniques, and response distributions by model and condition. Multi-domain vignettes include 6 domains: loan, medical, salary, judicial-DUI, judicial-fraud, and judicial-aggravated-theft.

References

Marcel Binz and Eric Schulz. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6):e2218523120, 2023. doi: 10.1073/pnas.2218523120.

Yifan Chen et al. Cognitive biases in LLM-assisted software development. *arXiv preprint arXiv:2601.08045*, 2025.

Leon Chlon, Sarah Rashidi, Zein Khamis, and MarcAntonio M. Awada. LLMs are Bayesian, in expectation, not in realization. *arXiv preprint arXiv:2507.11768*, 2025. doi: 10.48550/arXiv.2507.11768.

Birte Englich, Thomas Mussweiler, and Fritz Strack. Playing dice with criminal sentences: The influence of irrelevant anchors on experts' judicial decision making. *Personality and Social Psychology Bulletin*, 32(2):188–200, 2006. doi: 10.1177/0146167205282152.

Yucheng Huang et al. An empirical study of the anchoring effect in LLMs: Existence, mechanism, and potential mitigations. *arXiv preprint arXiv:2505.15392*, 2025.

Karen E Jacowitz and Daniel Kahneman. Measures of anchoring in estimation tasks. *Personality and Social Psychology Bulletin*, 21(11):1161–1166, 1995.

Erik Jones and Jacob Steinhardt. Capturing failures of large language models via human cognitive biases. *Advances in Neural Information Processing Systems*, 35:11785–11799, 2022.

Gary Klein. *The Power of Intuition: How to Use Your Gut Feelings to Make Better Decisions at Work*. Currency, 2007. ISBN 978-0385502894.

Kahee Lim et al. DeFrame: Debiasing large language models against framing effects. *arXiv preprint arXiv:2602.04306*, 2026.

Yifan Lyu et al. Self-adaptive cognitive debiasing for large language models. *arXiv preprint arXiv:2504.04141*, 2025.

Olivier Sibony. *You're About to Make a Terrible Mistake!: How Biases Distort Decision-Making and What You Can Do to Fight Them*. Little, Brown Spark, 2019. ISBN 978-0316494984.

Peiyang Song, Pengrui Han, and Noah Goodman. Large language model reasoning failures. *arXiv preprint arXiv:2602.06176*, 2026. TMLR 2026 Survey Certification.

Zailong Tian et al. Overconfidence in LLM-as-a-judge: Diagnosis and confidence-driven solution. *arXiv preprint arXiv:2508.06225*, 2025. doi: 10.48550/arXiv.2508.06225.

Amos Tversky and Daniel Kahneman. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131, 1974. doi: 10.1126/science.185.4157.1124.